

CART 470

Concordia Soundscape Explorer: Living Learning Contract

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Thesis

Project Focus

The main goals of Capstone Prototyping are emphasizing team-based collaboration and simulating the real-life scenario of working with a client - skills which we will hone as we develop the Concordia Soundscape project, proposed by Professor Florian Grond. Our primary focus is to produce an immersive experience of the Concordia campus through sound rather than just visuals, allowing more users - including those with visual impairments - to participate. It is important to develop this project modularly so that anyone can build and share their own perception of Concordia, thus making the prototype expandable.

Project Outcome

We intend to develop a modular web-based platform in which the user can interact with various 3D points of interest spread across a map - each fitted with dynamic soundscapes tailored to specific times of day. This sound will be captured using a multi-channel 3D surround audio experience, which will help accurately simulate the ambient, frequently hectic environment of our campus such that it remains accessible even to those with visual impairments. Each location will be accessible via a bird's-eye view of the map (similar to Google Maps or a standard GPS), as well as through floor plans laying out the Concordia campus.

Knowledge Base/Fields/Domains/Keywords

Modularity, immersive, sound, accessibility, 3-dimensional space, web development.

Goals

Learning Objectives

Since our client, Professor Florian Grond, works in a much different domain than many in our group, our primary goal is to work with him as closely as possible to ensure the project meets his expectations. To this end, much of our time will be dedicated to learning the software, equipment, and general skills required for this undertaking, as well as familiarizing ourselves with the fundamentals of a sound-based experience. For instance, our group will

begin by learning how to use the specialized recording devices Professor Grond will provide, as well as discovering tools to capture 3D visuals (whether that be a basic panorama or something more advanced). We will then figure out Eclipsa and/or IAMF to configure the audio before assembling an accessible website that will link all the pieces together. We must always keep our end users in mind to ensure our product is welcoming and functional to a wide variety of people.

To bridge the gap between our group's expertise and our client's, close communication will be imperative. Learning how to make the best use of our client's time, as well as our own, is something that will be at the forefront of our development process. Furthermore, communication with each other will be essential to ensuring everyone stays on the right track because, at the end of the day, twelve weeks will pass before we know it.

Learning Activities

- Playing with audio: Learn from Professor Grond about 3D and immersive audio, and get hands-on with different microphones, audio devices, and software to see what they can do.
- Building the map: Try different ways of putting a campus map into the project and connecting it to the sounds we record. Work together so everyone gets a chance to experiment with the different tools.
- Trying things out: Build small experiments with the map, sounds, and interactions, test them with the team, and keep the ideas that work best for the final prototype.

Outcome #1: Learn how to use different audio technologies, devices, and software to create an immersive sound experience.

Outcome #2: Learn how to work together and develop a pipeline while experimenting with new tools and finding creative solutions for the project.

Milestones

1. Implement Google Maps / Open Maps API into a web platform
2. Add 3D immersive audio to a web site or VR engine

3. Modularity and implementation of surround 360 view

Weekly Plan

Week #	Topic	Detail	Work Due (Deadlines)
Week 1 September 9	Introduction to CART 470	We will familiarize ourselves with the course and the potential projects	
Week 2 September 16	Preparing for client meeting	We will do some background research into the soundscape project and set up team communications	
Week 3 September 23	Preliminary research	We will research the base technologies required to build a website that integrates maps, immersive audio, and interactivity	
Week 4 September 30	Advanced research	We will gather as much information as possible on the technologies required and discussed with client	
Week 5 October 7	Initial data gathering	We will begin gathering audio data at various locations around campus using specialized equipment	
Week 6 October 21	Audio experiments	We will experiment with the different audio devices and software and start testing how the recordings could work in the project	
Week 7 October 28	Map prototype	We will start building the campus map and experiment with different ways of connecting locations to the sound experiences	
Week 8 November 4	Prototype development	We will combine the map, audio, and interactive elements into a first working version of the project	

Week 9 November 11	Testing and improvements	We will test the prototype together, find problems, and make changes based on what we learn	
Week 10 November 18	Final development	We will finish the main features and improve the overall experience based on our testing and feedback	A prototype with all features implemented and somewhat working, debugging will be needed
Week 11 November 25	Final Tests and preparation / Debugging	We will do final tests, fix any remaining problems, and prepare the project and presentation	Near-final prototype
Week 12 December 2	Presentation of the project	We will present the final project and show the process and results of our work	Final version of the prototype